ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 2 [10 marks]

Q2(a) [2 marks]

At equilibrium the velocity should be zero. The torque and gravity torque must balance.

Q2(b) [2 marks]

$$A = [[0, 1], [-mg\ell sin(60^\circ)/J, -b/J]] = [[0, 1], [-106.2, -2.5]],$$
$$B = [[0], [50]]$$

Q2(c) [4 marks]

At steady state the derivative terms vanish. The proportional torque is Kp(θd-θ), which must balance mgℓ sin θ.

Q2(d) [2 marks]

The added gravity torque should cancel the load, so the steady-state error becomes zero.

Question 3 [10 marks]

Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

Q3(b) [2 marks]

Tustin uses s = (2/T)(z-1)/(z+1).

Q3(c) [4 marks]

A-LC=[[-ℓ_1,1], [-ℓ_2,-2]]. I obtain p(s)=s^2+(ℓ_1+2)s+(ℓ_1+ℓ_2). Matching $(s+10)^2$ gives ℓ_1=18 and ℓ_2=82.

Q3(d) [2 marks]

The observer is better and the filter is safer in some cases.

Question 4 [10 marks]

Q4(a) [2 marks]

Q4(b) [2 marks]

Q4(c) [4 marks]

Q4(d) [2 marks]

Integral action removes steady-state error, although this only holds for the nominal values of b and c. Saturation can cause windup, which should be handled by anti-windup.